

Question 1

Write in index notation

$$(a) \quad \begin{cases} x_1 + c_{11}y_1 + c_{12}y_2 = d_{11}z_1 + d_{21}z_2, \\ x_2 + c_{21}y_1 + c_{22}y_2 = d_{12}z_1 + d_{22}z_2, \end{cases}$$

$$(b) \quad \begin{cases} Et_1 = s_1 + A_{11}v_1 + A_{22}v_2, \\ Et_2 = s_2 + A_{22}v_2 + A_{11}v_1. \end{cases}$$

Question 2

Write in full

$$(a) \quad m_\alpha = t_{\alpha\beta}n_\beta, \quad \alpha, \beta = 1, 2,$$

$$(b) \quad A_{rs} = m_r n_s + \delta_{rs}, \quad r, s = 1, 2.$$

Question 3

Let

$$u_\alpha = (4, 3), \quad v_\alpha = (2, -1), \quad w = 2,$$

and

$$A_{\alpha\beta} = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}.$$

Evaluate

$$(a) \quad u_\alpha A_{\alpha\beta}, \quad (b) \quad A_{\alpha\alpha} - w,$$

$$(c) \quad A_{\alpha\beta} - w\delta_{\alpha\beta}, \quad (d) \quad u_\lambda v_\lambda.$$

Question 4

Write in simplest form Assume that the index range is 2.

$$(a) \quad \delta_{\alpha\alpha}, \quad (b) \quad \delta_{\lambda\mu}\delta_{\lambda\mu}, \quad (c) \quad u_\alpha v_\beta \delta_{\alpha\beta}.$$

Question 5

If

$$A_{ij} = x_i x_j, \quad i, j = 1, 2,$$

evaluate

$$(a) \quad A_{ij,j}; \quad (b) \quad A_{ij,ij}.$$

Question 6

Given the second-order tensor

$$a_{ij} = \begin{pmatrix} 0 & -1 & 3 \\ 1 & 0 & 2 \\ -3 & -2 & 0 \end{pmatrix},$$

find the components of the tensor in the coordinate system $\underline{x}_i$ defined by

$$x_i = \alpha_{ij} \underline{x}_j,$$

where

$$\alpha_{ij} = \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Question 7

(a) For

$$A_{ij} = x_i^2 x_j + 2x_1 x_2 \delta_{ij}, \quad (i, j = 1, 2),$$

calculate

$$A_{ij,i} \quad \text{and} \quad A_{kl,kl}.$$

(b) Let

$$u_\alpha = (2, 1), \quad v_\alpha = (1, 5), \quad w = 6,$$

and

$$A_{\alpha\beta} = \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Evaluate

$$u_\alpha A_{\alpha\beta} \quad \text{and} \quad A_{\alpha\beta} - 3\delta_{\alpha\beta}.$$

Question 8

Compute

$$A_{pq,q} \quad \text{and} \quad A_{pq,pq},$$

where

$$A_{ij} = x_i x_j^3 + 5x_2 \delta_{ij}, \quad (i, j = 1, 2).$$

Question 9

Use the relation

$$\sigma_{ij} = -p\delta_{ij} + \mu(u_{i,j} + u_{j,i})$$

to simplify the expression

$$-pv_j n_j + \mu(u_{i,j} + u_{j,i}) v_i n_j.$$

Question 10

Given the vectors

$$\mathbf{A} = (1, 2, 3)$$

and

$$\mathbf{B} = (4, 5, 6),$$

use the Levi–Civita symbol

$$(\mathbf{A} \times \mathbf{B})_i = \varepsilon_{ijk} A_j B_k$$

to compute the cross product

$$\mathbf{A} \times \mathbf{B}.$$

Question 11

Given the matrix

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & 6 \end{bmatrix},$$

use the Levi–Civita symbol

$$\det(A) = \varepsilon_{ijk} A_{1i} A_{2j} A_{3k}$$

to calculate the determinant of the matrix.

Question 12

Consider the vector field

$$\mathbf{A}(x, y, z) = (xy, yz, zx).$$

Using the Levi–Civita symbol,

$$(\nabla \times \mathbf{A})_i = \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j},$$

find the curl of the vector field,

$$\nabla \times \mathbf{A}.$$

Hint: Compute each component of the curl separately by evaluating the required partial derivatives and using the values of the Levi–Civita symbol.